SG No. 135 D_{4h}^{13} $P4_2/mbc$ [tetragonal]

* plus set: $+[0, 0, 0]$

* Wyckoff site: `4a`, site symmetry: `2/m..`

Table 1: Wyckoff bond: `4a@4a`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[0, 0, 0]$	`[1,-2,-9,10]`
2	$[-Y, X, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[3,-4,-11,12]`
3	$[-X, Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[5,-6,-13,14]`
4	$[Y, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[7,-8,-15,16]`

Table 2: Wyckoff bond: `4b@4a`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[0, 0, Z]$	$[0, 0, 0]$	`[1,2,-9,-10]`
2	$[0, 0, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[3,4,-11,-12]`
3	$[0, 0, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[5,6,-13,-14]`
4	$[0, 0, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[7,8,-15,-16]`

Table 3: Wyckoff bond: `8c@4a`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, 0]$	`[1,-9]`
2	$[-X, -Y, Z]$	$[0, 0, 0]$	`[2,-10]`
3	$[-Y, X, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[3,-11]`
4	$[Y, -X, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[4,-12]`
5	$[-X, Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[5,-13]`
6	$[X, -Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[6,-14]`
7	$[Y, X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[7,-15]`
8	$[-Y, -X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[8,-16]`

* Wyckoff site: `4b`, site symmetry: `-4..`

Table 4: Wyckoff bond: `4a@4b`, bond symmetry: `-4..`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[0, 0, \frac{1}{4}\right]$	`[1,2,-11,-12]`
2	$[0, 0, Z]$	$\left[0, 0, \frac{3}{4}\right]$	`[3,4,-9,-10]`
3	$[0, 0, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[5,6,-15,-16]`
4	$[0, 0, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[7,8,-13,-14]`

Table 5: Wyckoff bond: `8b@4b`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[0, 0, \frac{1}{4}\right]$	`[1,-2]`

continued ...

Table 5

No.	vector	center	mapping
2	$[-Y, X, 0]$	$\left[0, 0, \frac{3}{4}\right]$	`[3,-4]`
3	$[-X, Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[5,-6]`
4	$[Y, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[7,-8]`
5	$[-X, -Y, 0]$	$\left[0, 0, \frac{3}{4}\right]$	`[9,-10]`
6	$[Y, -X, 0]$	$\left[0, 0, \frac{1}{4}\right]$	`[11,-12]`
7	$[X, -Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[13,-14]`
8	$[-Y, -X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[15,-16]`

Table 6: Wyckoff bond: `16c@4b`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, 0, \frac{1}{4}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[0, 0, \frac{1}{4}\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[0, 0, \frac{3}{4}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[0, 0, \frac{3}{4}\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[8]`
9	$[-X, -Y, -Z]$	$\left[0, 0, \frac{3}{4}\right]$	`[9]`
10	$[X, Y, -Z]$	$\left[0, 0, \frac{3}{4}\right]$	`[10]`
11	$[Y, -X, -Z]$	$\left[0, 0, \frac{1}{4}\right]$	`[11]`
12	$[-Y, X, -Z]$	$\left[0, 0, \frac{1}{4}\right]$	`[12]`
13	$[X, -Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[13]`
14	$[-X, Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[14]`
15	$[-Y, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[15]`
16	$[Y, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[16]`

* Wyckoff site: `4c`, site symmetry: `2/m..`

Table 7: Wyckoff bond: `4a@4c`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[0, \frac{1}{2}, 0\right]$	`[1,-2,-9,10]`
2	$[-Y, X, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[3,-4,-11,12]`
3	$[-X, Y, 0]$	$\left[\frac{1}{2}, 0, 0\right]$	`[5,-6,-13,14]`
4	$[Y, X, 0]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[7,-8,-15,16]`

Table 8: Wyckoff bond: `4b@4c`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[1,2,-9,-10]`
2	$[0, 0, Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[3,4,-11,-12]`
3	$[0, 0, -Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[5,6,-13,-14]`

continued ...

Table 8

No.	vector	center	mapping
4	$[0, 0, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[7,8,-15,-16]`

Table 9: Wyckoff bond: `8c@4c`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[1,-9]`
2	$[-X, -Y, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[2,-10]`
3	$[-Y, X, Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[3,-11]`
4	$[Y, -X, Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[4,-12]`
5	$[-X, Y, -Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[5,-13]`
6	$[X, -Y, -Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[6,-14]`
7	$[Y, X, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[7,-15]`
8	$[-Y, -X, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[8,-16]`

* Wyckoff site: `4d`, site symmetry: `2.22`

Table 10: Wyckoff bond: `4a@4d`, bond symmetry: `2.22`

No.	vector	center	mapping
1	$[X, X, 0]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[1,-2,7,-8]`
2	$[-X, X, 0]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[3,-4,5,-6]`
3	$[-X, -X, 0]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[9,-10,15,-16]`
4	$[X, -X, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[11,-12,13,-14]`

Table 11: Wyckoff bond: `4b@4d`, bond symmetry: `2.22`

No.	vector	center	mapping
1	$[X, -X, 0]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[1,-2,-7,8]`
2	$[X, X, 0]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[3,-4,-5,6]`
3	$[-X, X, 0]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[9,-10,-15,16]`
4	$[-X, -X, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[11,-12,-13,14]`

Table 12: Wyckoff bond: `4c@4d`, bond symmetry: `2.22`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[1,2,-7,-8]`
2	$[0, 0, Z]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[3,4,-5,-6]`
3	$[0, 0, -Z]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[9,10,-15,-16]`
4	$[0, 0, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[11,12,-13,-14]`

Table 13: Wyckoff bond: `8d@4d`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, X, Z]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[1,-8]`
2	$[-X, -X, Z]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[2,-7]`
3	$[-X, X, Z]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[3,-6]`
4	$[X, -X, Z]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[4,-5]`
5	$[-X, -X, -Z]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[9,-16]`
6	$[X, X, -Z]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[10,-15]`
7	$[X, -X, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[11,-14]`
8	$[-X, X, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[12,-13]`

Table 14: Wyckoff bond: `8e@4d`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, -X, Z]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[1,-7]`
2	$[-X, X, Z]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[2,-8]`
3	$[X, X, Z]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[3,-5]`
4	$[-X, -X, Z]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[4,-6]`
5	$[-X, X, -Z]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[9,-15]`
6	$[X, -X, -Z]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[10,-16]`
7	$[-X, -X, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[11,-13]`
8	$[X, X, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[12,-14]`

Table 15: Wyckoff bond: `8f@4d`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[1,-2]`
2	$[-Y, X, 0]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[3,-4]`
3	$[-X, Y, 0]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[5,-6]`
4	$[Y, X, 0]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[7,-8]`
5	$[-X, -Y, 0]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[9,-10]`
6	$[Y, -X, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[11,-12]`
7	$[X, -Y, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[13,-14]`
8	$[-Y, -X, 0]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[15,-16]`

Table 16: Wyckoff bond: `16g@4d`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[7]`

continued ...

Table 16

No.	vector	center	mapping
8	$[-Y, -X, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[8]`
9	$[-X, -Y, -Z]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[9]`
10	$[X, Y, -Z]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[10]`
11	$[Y, -X, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[11]`
12	$[-Y, X, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[12]`
13	$[X, -Y, Z]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[13]`
14	$[-X, Y, Z]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[14]`
15	$[-Y, -X, Z]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[15]`
16	$[Y, X, Z]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[16]`

* Wyckoff site: `8e`, site symmetry: `2..`

Table 17: Wyckoff bond: `8a@8e`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[0, 0, z]$	`[1,-2]`
2	$[-Y, X, 0]$	$\left[0, 0, z + \frac{1}{2}\right]$	`[3,-4]`
3	$[-X, Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[5,-6]`
4	$[Y, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2} - z\right]$	`[7,-8]`
5	$[-X, -Y, 0]$	$[0, 0, -z]$	`[9,-10]`
6	$[Y, -X, 0]$	$\left[0, 0, \frac{1}{2} - z\right]$	`[11,-12]`
7	$[X, -Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[13,-14]`
8	$[-Y, -X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, z + \frac{1}{2}\right]$	`[15,-16]`

Table 18: Wyckoff bond: `8b@8e`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, 0, Z]$	$[0, 0, z]$	`[1,2]`
2	$[0, 0, Z]$	$\left[0, 0, z + \frac{1}{2}\right]$	`[3,4]`
3	$[0, 0, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[5,6]`
4	$[0, 0, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2} - z\right]$	`[7,8]`
5	$[0, 0, -Z]$	$[0, 0, -z]$	`[9,10]`
6	$[0, 0, -Z]$	$\left[0, 0, \frac{1}{2} - z\right]$	`[11,12]`
7	$[0, 0, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[13,14]`
8	$[0, 0, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z + \frac{1}{2}\right]$	`[15,16]`

Table 19: Wyckoff bond: `16c@8e`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, z]$	`[1]`
2	$[-X, -Y, Z]$	$[0, 0, z]$	`[2]`
3	$[-Y, X, Z]$	$\left[0, 0, z + \frac{1}{2}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[0, 0, z + \frac{1}{2}\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[5]`

continued ...

Table 19

No.	vector	center	mapping
6	$[X, -Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2} - z\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2} - z\right]$	`[8]`
9	$[-X, -Y, -Z]$	$[0, 0, -z]$	`[9]`
10	$[X, Y, -Z]$	$[0, 0, -z]$	`[10]`
11	$[Y, -X, -Z]$	$\left[0, 0, \frac{1}{2} - z\right]$	`[11]`
12	$[-Y, X, -Z]$	$\left[0, 0, \frac{1}{2} - z\right]$	`[12]`
13	$[X, -Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[13]`
14	$[-X, Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[14]`
15	$[-Y, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z + \frac{1}{2}\right]$	`[15]`
16	$[Y, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z + \frac{1}{2}\right]$	`[16]`

* Wyckoff site: `8f`, site symmetry: `2..`

Table 20: Wyckoff bond: `8a@8f`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[0, \frac{1}{2}, z\right]$	`[1,-2]`
2	$[-Y, X, 0]$	$\left[\frac{1}{2}, 0, z + \frac{1}{2}\right]$	`[3,-4]`
3	$[-X, Y, 0]$	$\left[\frac{1}{2}, 0, -z\right]$	`[5,-6]`
4	$[Y, X, 0]$	$\left[0, \frac{1}{2}, \frac{1}{2} - z\right]$	`[7,-8]`
5	$[-X, -Y, 0]$	$\left[0, \frac{1}{2}, -z\right]$	`[9,-10]`
6	$[Y, -X, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{2} - z\right]$	`[11,-12]`
7	$[X, -Y, 0]$	$\left[\frac{1}{2}, 0, z\right]$	`[13,-14]`
8	$[-Y, -X, 0]$	$\left[0, \frac{1}{2}, z + \frac{1}{2}\right]$	`[15,-16]`

Table 21: Wyckoff bond: `8b@8f`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[0, \frac{1}{2}, z\right]$	`[1,2]`
2	$[0, 0, Z]$	$\left[\frac{1}{2}, 0, z + \frac{1}{2}\right]$	`[3,4]`
3	$[0, 0, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	`[5,6]`
4	$[0, 0, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{2} - z\right]$	`[7,8]`
5	$[0, 0, -Z]$	$\left[0, \frac{1}{2}, -z\right]$	`[9,10]`
6	$[0, 0, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2} - z\right]$	`[11,12]`
7	$[0, 0, Z]$	$\left[\frac{1}{2}, 0, z\right]$	`[13,14]`
8	$[0, 0, Z]$	$\left[0, \frac{1}{2}, z + \frac{1}{2}\right]$	`[15,16]`

Table 22: Wyckoff bond: `16c@8f`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, \frac{1}{2}, z\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[0, \frac{1}{2}, z\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[\frac{1}{2}, 0, z + \frac{1}{2}\right]$	`[3]`

continued ...

Table 22

No.	vector	center	mapping
4	$[Y, -X, Z]$	$\left[\frac{1}{2}, 0, z+\frac{1}{2}\right]$	[4]
5	$[-X, Y, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	[5]
6	$[X, -Y, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	[6]
7	$[Y, X, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}-z\right]$	[7]
8	$[-Y, -X, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}-z\right]$	[8]
9	$[-X, -Y, -Z]$	$\left[0, \frac{1}{2}, -z\right]$	[9]
10	$[X, Y, -Z]$	$\left[0, \frac{1}{2}, -z\right]$	[10]
11	$[Y, -X, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}-z\right]$	[11]
12	$[-Y, X, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}-z\right]$	[12]
13	$[X, -Y, Z]$	$\left[\frac{1}{2}, 0, z\right]$	[13]
14	$[-X, Y, Z]$	$\left[\frac{1}{2}, 0, z\right]$	[14]
15	$[-Y, -X, Z]$	$\left[0, \frac{1}{2}, z+\frac{1}{2}\right]$	[15]
16	$[Y, X, Z]$	$\left[0, \frac{1}{2}, z+\frac{1}{2}\right]$	[16]

* Wyckoff site: 8g, site symmetry: ..2

Table 23: Wyckoff bond: 8a@8g, bond symmetry: ..2

No.	vector	center	mapping
1	$[X, -X, Z]$	$\left[x, x+\frac{1}{2}, \frac{1}{4}\right]$	[1,-7]
2	$[-X, X, Z]$	$\left[-x, \frac{1}{2}-x, \frac{1}{4}\right]$	[2,-8]
3	$[X, X, Z]$	$\left[\frac{1}{2}-x, x, \frac{3}{4}\right]$	[3,-5]
4	$[-X, -X, Z]$	$\left[x+\frac{1}{2}, -x, \frac{3}{4}\right]$	[4,-6]
5	$[-X, X, -Z]$	$\left[-x, \frac{1}{2}-x, \frac{3}{4}\right]$	[9,-15]
6	$[X, -X, -Z]$	$\left[x, x+\frac{1}{2}, \frac{3}{4}\right]$	[10,-16]
7	$[-X, -X, -Z]$	$\left[x+\frac{1}{2}, -x, \frac{1}{4}\right]$	[11,-13]
8	$[X, X, -Z]$	$\left[\frac{1}{2}-x, x, \frac{1}{4}\right]$	[12,-14]

Table 24: Wyckoff bond: 8b@8g, bond symmetry: ..2

No.	vector	center	mapping
1	$[X, X, 0]$	$\left[x, x+\frac{1}{2}, \frac{1}{4}\right]$	[1,7]
2	$[-X, -X, 0]$	$\left[-x, \frac{1}{2}-x, \frac{1}{4}\right]$	[2,8]
3	$[-X, X, 0]$	$\left[\frac{1}{2}-x, x, \frac{3}{4}\right]$	[3,5]
4	$[X, -X, 0]$	$\left[x+\frac{1}{2}, -x, \frac{3}{4}\right]$	[4,6]
5	$[-X, -X, 0]$	$\left[-x, \frac{1}{2}-x, \frac{3}{4}\right]$	[9,15]
6	$[X, X, 0]$	$\left[x, x+\frac{1}{2}, \frac{3}{4}\right]$	[10,16]
7	$[X, -X, 0]$	$\left[x+\frac{1}{2}, -x, \frac{1}{4}\right]$	[11,13]
8	$[-X, X, 0]$	$\left[\frac{1}{2}-x, x, \frac{1}{4}\right]$	[12,14]

Table 25: Wyckoff bond: 16c@8g, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, x+\frac{1}{2}, \frac{1}{4}\right]$	[1]

continued ...

Table 25

No.	vector	center	mapping
2	$[-X, -Y, Z]$	$\left[-x, \frac{1}{2}-x, \frac{1}{4}\right]$	[2]
3	$[-Y, X, Z]$	$\left[\frac{1}{2}-x, x, \frac{3}{4}\right]$	[3]
4	$[Y, -X, Z]$	$\left[x+\frac{1}{2}, -x, \frac{3}{4}\right]$	[4]
5	$[-X, Y, -Z]$	$\left[\frac{1}{2}-x, x, \frac{3}{4}\right]$	[5]
6	$[X, -Y, -Z]$	$\left[x+\frac{1}{2}, -x, \frac{3}{4}\right]$	[6]
7	$[Y, X, -Z]$	$\left[x, x+\frac{1}{2}, \frac{1}{4}\right]$	[7]
8	$[-Y, -X, -Z]$	$\left[-x, \frac{1}{2}-x, \frac{1}{4}\right]$	[8]
9	$[-X, -Y, -Z]$	$\left[-x, \frac{1}{2}-x, \frac{3}{4}\right]$	[9]
10	$[X, Y, -Z]$	$\left[x, x+\frac{1}{2}, \frac{3}{4}\right]$	[10]
11	$[Y, -X, -Z]$	$\left[x+\frac{1}{2}, -x, \frac{1}{4}\right]$	[11]
12	$[-Y, X, -Z]$	$\left[\frac{1}{2}-x, x, \frac{1}{4}\right]$	[12]
13	$[X, -Y, Z]$	$\left[x+\frac{1}{2}, -x, \frac{1}{4}\right]$	[13]
14	$[-X, Y, Z]$	$\left[\frac{1}{2}-x, x, \frac{1}{4}\right]$	[14]
15	$[-Y, -X, Z]$	$\left[-x, \frac{1}{2}-x, \frac{3}{4}\right]$	[15]
16	$[Y, X, Z]$	$\left[x, x+\frac{1}{2}, \frac{3}{4}\right]$	[16]

* Wyckoff site: 8h, site symmetry: m..

Table 26: Wyckoff bond: 8a@8h, bond symmetry: m..

No.	vector	center	mapping
1	$[X, Y, 0]$	$[x, y, 0]$	[1,10]
2	$[-X, -Y, 0]$	$[-x, -y, 0]$	[2,9]
3	$[-Y, X, 0]$	$\left[-y, x, \frac{1}{2}\right]$	[3,12]
4	$[Y, -X, 0]$	$\left[y, -x, \frac{1}{2}\right]$	[4,11]
5	$[-X, Y, 0]$	$\left[\frac{1}{2}-x, y+\frac{1}{2}, 0\right]$	[5,14]
6	$[X, -Y, 0]$	$\left[x+\frac{1}{2}, \frac{1}{2}-y, 0\right]$	[6,13]
7	$[Y, X, 0]$	$\left[y+\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}\right]$	[7,16]
8	$[-Y, -X, 0]$	$\left[\frac{1}{2}-y, \frac{1}{2}-x, \frac{1}{2}\right]$	[8,15]

Table 27: Wyckoff bond: 8b@8h, bond symmetry: m..

No.	vector	center	mapping
1	$[0, 0, Z]$	$[x, y, 0]$	[1,-10]
2	$[0, 0, Z]$	$[-x, -y, 0]$	[2,-9]
3	$[0, 0, Z]$	$\left[-y, x, \frac{1}{2}\right]$	[3,-12]
4	$[0, 0, Z]$	$\left[y, -x, \frac{1}{2}\right]$	[4,-11]
5	$[0, 0, -Z]$	$\left[\frac{1}{2}-x, y+\frac{1}{2}, 0\right]$	[5,-14]
6	$[0, 0, -Z]$	$\left[x+\frac{1}{2}, \frac{1}{2}-y, 0\right]$	[6,-13]
7	$[0, 0, -Z]$	$\left[y+\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}\right]$	[7,-16]
8	$[0, 0, -Z]$	$\left[\frac{1}{2}-y, \frac{1}{2}-x, \frac{1}{2}\right]$	[8,-15]

Table 28: Wyckoff bond: `16c@8h`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, 0]$	`[1]`
2	$[-X, -Y, Z]$	$[-x, -y, 0]$	`[2]`
3	$[-Y, X, Z]$	$\left[-y, x, \frac{1}{2}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[y, -x, \frac{1}{2}\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[\frac{1}{2} - x, y + \frac{1}{2}, 0\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[x + \frac{1}{2}, \frac{1}{2} - y, 0\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[y + \frac{1}{2}, x + \frac{1}{2}, \frac{1}{2}\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[\frac{1}{2} - y, \frac{1}{2} - x, \frac{1}{2}\right]$	`[8]`
9	$[-X, -Y, -Z]$	$[-x, -y, 0]$	`[9]`
10	$[X, Y, -Z]$	$[x, y, 0]$	`[10]`
11	$[Y, -X, -Z]$	$\left[y, -x, \frac{1}{2}\right]$	`[11]`
12	$[-Y, X, -Z]$	$\left[-y, x, \frac{1}{2}\right]$	`[12]`
13	$[X, -Y, Z]$	$\left[x + \frac{1}{2}, \frac{1}{2} - y, 0\right]$	`[13]`
14	$[-X, Y, Z]$	$\left[\frac{1}{2} - x, y + \frac{1}{2}, 0\right]$	`[14]`
15	$[-Y, -X, Z]$	$\left[\frac{1}{2} - y, \frac{1}{2} - x, \frac{1}{2}\right]$	`[15]`
16	$[Y, X, Z]$	$\left[y + \frac{1}{2}, x + \frac{1}{2}, \frac{1}{2}\right]$	`[16]`

* Wyckoff site: `16i`, site symmetry: `1`

Table 29: Wyckoff bond: `16a@16i`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, z]$	`[1]`
2	$[-X, -Y, Z]$	$[-x, -y, z]$	`[2]`
3	$[-Y, X, Z]$	$\left[-y, x, z + \frac{1}{2}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[y, -x, z + \frac{1}{2}\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[\frac{1}{2} - x, y + \frac{1}{2}, -z\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[x + \frac{1}{2}, \frac{1}{2} - y, -z\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[y + \frac{1}{2}, x + \frac{1}{2}, \frac{1}{2} - z\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[\frac{1}{2} - y, \frac{1}{2} - x, \frac{1}{2} - z\right]$	`[8]`
9	$[-X, -Y, -Z]$	$[-x, -y, -z]$	`[9]`
10	$[X, Y, -Z]$	$[x, y, -z]$	`[10]`
11	$[Y, -X, -Z]$	$\left[y, -x, \frac{1}{2} - z\right]$	`[11]`
12	$[-Y, X, -Z]$	$\left[-y, x, \frac{1}{2} - z\right]$	`[12]`
13	$[X, -Y, Z]$	$\left[x + \frac{1}{2}, \frac{1}{2} - y, z\right]$	`[13]`
14	$[-X, Y, Z]$	$\left[\frac{1}{2} - x, y + \frac{1}{2}, z\right]$	`[14]`
15	$[-Y, -X, Z]$	$\left[\frac{1}{2} - y, \frac{1}{2} - x, z + \frac{1}{2}\right]$	`[15]`
16	$[Y, X, Z]$	$\left[y + \frac{1}{2}, x + \frac{1}{2}, z + \frac{1}{2}\right]$	`[16]`